

Revise: Linear Algebra Primer

Vectors and Matrices

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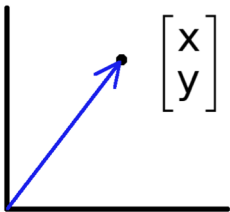
- Basic Matrix Operations
- Special Matrices

In computer vision/Image Processing: vectors and matrices are collections of ordered numbers that represent:

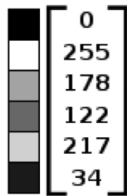
- movements in space, scaling factors
- pixel brightnesses
-

- A column vector $v \in R^{n \times 1}$ where $v = \begin{pmatrix} v_1 \\ v_2 \\ \cdot \\ \cdot \\ \cdot \\ v_n \end{pmatrix}$
- A row vector $v^T \in R^{1 \times n}$: $v^T = [v_1, v_2, \dots, v_n]$ in which T is transpose operation

Vectors have two main uses



- Vectors can represent an offset in 2D or 3D space
- Points are just vectors from the origin



- Data (pixels, gradients at an image keypoint, etc) can also be treated as a vector
- Such vectors don't have a geometric interpretation, but calculations like "distance" can still have value

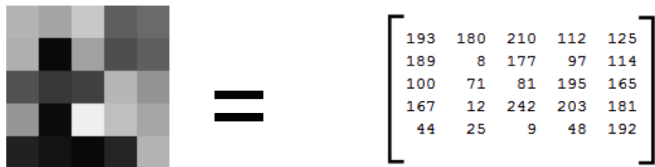
- A matrix $A \in R^{m \times n}$ is an array of numbers with size m rows

and n columns $A = \begin{pmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ \dots & & & & \\ a_{m1} & a_{m2} & a_{m3} & \dots & a_{mn} \end{pmatrix}$

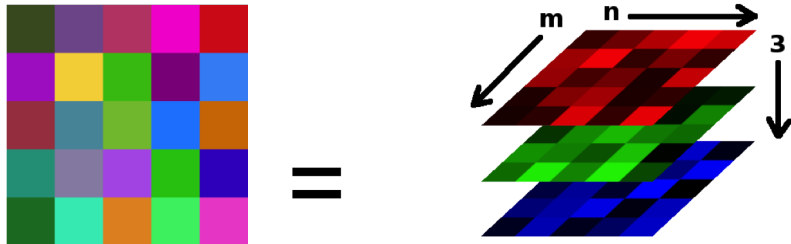
- If $m = n$, we say that A is square

Images

- Gray images have one number per pixel, and are stored as an $m \times n$ matrix



- Color images have 3 numbers per pixel – red, green, and blue brightnesses. Stored as an $m \times n \times 3$ matrix



Basic Matrix Operations

- Addition
- Scaling
- Dot product
- Multiplication
- Transpose
- Inverse
- Determinant

Matrix Operations

■ Addition

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} + \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} a+1 & b+2 \\ c+3 & d+4 \end{bmatrix}$$

Can only add a matrix with matching dimensions, or a scalar.

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} + 7 = \begin{bmatrix} a+7 & b+7 \\ c+7 & d+7 \end{bmatrix}$$

■ Scaling

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \times 3 = \begin{bmatrix} 3a & 3b \\ 3c & 3d \end{bmatrix}$$

Matrix Operations

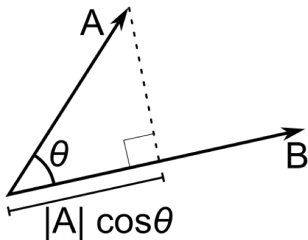
■ Inner product (dot product) of vectors

- Multiply corresponding entries of two vectors and add up the result

- $\mathbf{x} \cdot \mathbf{y}$ is also $|\mathbf{x}||\mathbf{y}| \cos(\text{the angle between } \mathbf{x} \text{ and } \mathbf{y})$

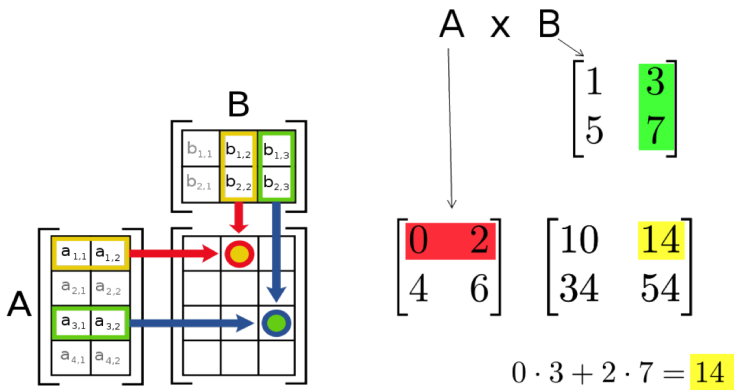
$$\mathbf{x}^T \mathbf{y} = \begin{bmatrix} x_1 & \dots & x_n \end{bmatrix} \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} = \sum_{i=1}^n x_i y_i \quad (\text{scalar})$$

- If B is a unit vector, then $\mathbf{A} \cdot \mathbf{B}$ gives the length of A which lies in the direction of B



Matrix Operations: Multiplication

- The product AB is:



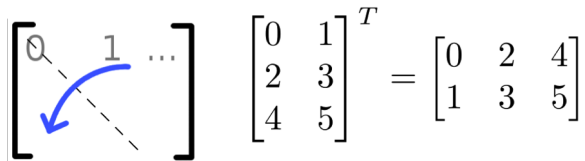
- Each entry in the result is (that row of A) dot product with (that column of B)
- Many uses, which will be covered later

- Powers

- By convention, we can refer to the matrix product AA as A^2 , and AAA as A^3 , etc.
- Obviously only square matrices can be multiplied that way

Matrix Operations

- Transpose – flip matrix, so row 1 becomes column 1


$$\begin{bmatrix} 0 & 1 & \dots \end{bmatrix}^T = \begin{bmatrix} 0 & 1 \\ 2 & 3 \\ 4 & 5 \end{bmatrix} = \begin{bmatrix} 0 & 2 & 4 \\ 1 & 3 & 5 \end{bmatrix}$$

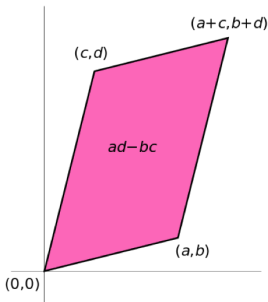
- A useful identity:

$$(ABC)^T = C^T B^T A^T$$

Matrix Operations

■ Determinant

- $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ then $\det(A) = ad - bc$ returns a scalar
- Represents area (or volume) of the parallelogram described by the vectors in the rows of the matrix



$$\det(\mathbf{AB}) = \det(\mathbf{BA})$$

$$\det(\mathbf{A}^{-1}) = \frac{1}{\det(\mathbf{A})}$$

$$\det(\mathbf{A}^T) = \det(\mathbf{A})$$

$$\det(\mathbf{A}) = 0 \Leftrightarrow \mathbf{A} \text{ is singular}$$

- Trace $\text{tr}(A)$ = sum of diagonal elements

$$\text{tr}\left(\begin{pmatrix} 1 & 3 \\ 5 & 7 \end{pmatrix}\right) = 1 + 7 = 8$$

- Invariant to a lot of transformations
- Properties:

$$\text{tr}(AB) = \text{tr}(BA) \tag{1}$$

$$\text{tr}(A + B) = \text{tr}(A) + \text{tr}(B) \tag{2}$$

Special Matrices

- Identity matrix I

- Square matrix, 1's along diagonal, 0's elsewhere

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

- $I \cdot [\text{another matrix}] = [\text{that matrix}]$

- Diagonal matrix

- Square matrix with numbers along diagonal, 0's elsewhere

$$\begin{bmatrix} 3 & 0 & 0 \\ 0 & 7 & 0 \\ 0 & 0 & 2.5 \end{bmatrix}$$

- A diagonal $\cdot$ [another matrix] scales the rows of that matrix

- Symmetric matrix: $A^T = A$

$$\begin{bmatrix} 1 & 2 & 5 \\ 2 & 1 & 7 \\ 5 & 7 & 1 \end{bmatrix}$$

- Skew-symmetric matrix $A^T = -A$

$$\begin{bmatrix} 1 & -2 & -5 \\ 2 & 1 & -7 \\ 5 & 7 & 1 \end{bmatrix}$$